

ASSIGNMENTS – OOADI

LECTURE 2

1. Make a use-case diagram for a small robot system that cleans garbage in a canteen. You do not need to have everything completed.
 2. Identify main modules that can realize the use-cases.
 3. Select one or a few modules and subdivide these modules.
 4. Describe how those would need to interact.
 5. Describe what entities are involved and for each entity, describe what we need to be able to do with them.
 6. Make (simple) code that demonstrates a selected part of the interactions you defined in step 4.
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7. Based on the below description, define a set of classes for things on a bicycle down to nuts and bolts. Define methods and attributes for each – you don't have to overdo this. Just so you have enough to roughly model a bicycle and be able to "use" each class in a limited way. I.e. use your senses to scope yourself.
8. Implement the main classes from your initial design. Add some simple functionality to key method that simply prints its functionality to the screen, e.g. "changing gear", "wheel off".
9. Create a bicycle class, that initiates the previous defined classes. Also add attributes and methods to the bicycle as you would use a normal bicycle.
10. Create a main() that initiates the bicycle and show that you can execute via code, the below description.

"With my Bicycle I can ride it forward by stepping on the pedals, and by changing gears I can go faster if I choose a higher gearing, but it also becomes harder to step on the pedals. For that reason, I start in first gear. I have front and backlights which switch on if it becomes dark. The wheels I can take off when I need to fix holes in the tire. When I brake, I can use both front and rear brakes."